

Project Proposal: Distributed Real-Time Air Quality Forecasting and Proactive Industrial Response Using IoT Streams

ENGR 5785G: REAL-TIME DATA ANALYTICS FOR IoT | PROJECT PROPOSAL
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1. Background and Project Scope

Industrial activities such as metal processing, chemical manufacturing, and surface treatment facilities are significant sources of air pollution in urban and peri-urban environments. These sectors release hazardous pollutants including nitrogen dioxide (NO_2), sulfur dioxide (SO_2), particulate matter ($PM_{2.5}$ and PM_{10}), and in some cases toxic compounds such as hexavalent chromium. Prolonged exposure to these pollutants is strongly associated with respiratory illness, cardiovascular disease, and long-term occupational health risks.

Despite advancements in environmental monitoring, most existing industrial air quality systems remain fundamentally reactive. Traditional monitoring infrastructure relies on fixed-location stations that provide periodic or delayed measurements, often requiring post-processing or laboratory validation. As a result, interventions occur after harmful concentrations have already been reached.

Recent developments in Internet of Things (IoT) sensing systems now enable continuous, high-frequency measurement of air quality and meteorological conditions. However, the key challenge is no longer data collection, but rather the transformation of heterogeneous, noisy, and distributed sensor streams into reliable real-time predictions that can support proactive decision-making within strict latency constraints.

This project proposes a distributed real-time air quality forecasting system that integrates IoT sensor streams, meteorological data, and spatiotemporal machine learning models to enable early prediction of hazardous pollution events and support automated industrial response strategies.

1.1 Geographic Scope

- **Primary Study area:** Industrial and urban regions covered by the OpenAQ real-time air quality network, with emphasis on areas containing dense industrial activity and mixed residential exposure.
- **Secondary Validation:** Selected Canadian urban monitoring locations using Environment Canada air quality and weather stations to evaluate environmental generalization under different climatic conditions.
- **Sensor coverage:** Distributed network of heterogeneous monitoring stations providing pollutant and meteorological measurements at varying spatial densities.

1.2 Temporal and Streaming Scope

- **Historical Training:** 2022–2024 archived air quality and weather observations.
- **Real-Time Target:** Sub-minute end-to-end latency from sensor ingestion to prediction output.
- **Forecast Horizons:** 5-minute, 30-minute, and 60-minute ahead pollutant concentration predictions.
- **Data frequency:** Near real-time streaming ingestion with updates ranging from 1-60 seconds depending on data source.

1.3 IoT and Analytics Scope

- **Sensor Modalities:** $PM_{2.5}$, PM_{10} , CO_2 , NO_2 , temperature, humidity, wind speed, and wind direction.
- **Stream Processing:** Apache Kafka for distributed ingestion and Apache Flink for real-time windowed computation.
- **Spatiotemporal modeling:** Graph-based neural networks for learning spatial diffusion patterns of pollutants across monitoring locations.
- **System output:** Real-time air quality forecasting API and industrial alert dashboard with predictive warnings.

2. Research Gaps

Although significant progress has been made in air quality forecasting and IoT-based environmental monitoring, several important limitations remain unresolved in current literature and industrial systems.

2.1 Lack of Streaming-Aware Air Quality Forecasting Systems: Most existing air quality prediction models are developed using static, offline datasets where data is pre-collected, cleaned, and batch processed. While these approaches achieve strong predictive performance, they fail to address real-world operational constraints such as continuous data arrival, delayed sensor updates, and irregular sampling intervals. As a result, the transition from offline models to real-time deployment remains largely unexplored.

2.2 Limited Modeling of Spatial Pollutant Diffusion in Industrial Zones: Current approaches often treat monitoring stations as independent time series or assume simplified spatial relationships. However, pollutant dispersion in industrial environments is inherently spatial and influenced by physical factors such as wind direction, emission source intensity, and geographical

barriers. Existing models rarely incorporate dynamic spatial dependencies that reflect real atmospheric diffusion behavior.

2.3 Sensor Noise, Drift, and Missing Data in Real Deployments: In real-world IoT deployments, air quality sensors frequently suffer from calibration drift, temporary outages, and communication failures. Most published work assumes clean and complete datasets, with limited attention to robust real-time imputation strategies that preserve both temporal continuity and spatial consistency.

2.4 Limited Integration of Forecasting with Proactive Control Systems: While many studies focus on predicting pollutant concentrations, relatively few extend these predictions into actionable systems that trigger automated responses. In industrial environments, early prediction alone is insufficient unless it is coupled with decision-making mechanisms such as ventilation activation or emission reduction protocols.

3. Research Objectives

This project aims to address the above gaps through the following objectives:

1. Design and implement a real-time IoT data streaming pipeline using Kafka and Flink for ingestion, processing, and feature extraction of heterogeneous environmental sensor data.
2. Develop a spatiotemporal graph-based forecasting model that captures dynamic relationships between monitoring stations and learns pollutant diffusion patterns across space and time.
3. Implement a robust sensor fault-tolerance and imputation module capable of handling missing, delayed, or corrupted sensor readings in real time.
4. Integrate forecasting outputs into a proactive industrial response simulation system capable of triggering early warnings and mitigation actions.
5. Evaluate the system under realistic streaming conditions and validate performance using multiple real-world air quality datasets.

4. Potential Methodology

The proposed system is structured into five main phases following a real-time IoT analytics pipeline: Ingest → Process → Model → Recover → Deploy.

4.1 Phase 1: IoT Stream Ingestion Layer: A distributed Kafka-based ingestion system will be deployed to handle multiple real-time data sources:

- OpenAQ real-time air quality API.

- Environment Canada meteorological feeds.
- Supplementary weather and environmental sensor streams.

Each data source will be assigned a dedicated Kafka topic with schema validation using Apache Avro to ensure consistency and reliability across heterogeneous inputs.

4.2 Phase 2: Real-Time Feature Engineering: Apache Flink will be used to process streaming environmental data with low latency and to generate enriched spatiotemporal features for downstream forecasting.

In addition to standard rolling statistics and lag features, this project explicitly incorporates context-aware spatial weighting driven by meteorological conditions, particularly wind direction and wind speed.

A key extension is the formulation of a direction-aware dynamic adjacency matrix:

$$G = (V, E, A(t))$$

where the time-varying adjacency $A(t)$ is not only a function of physical distance but is also modulated by atmospheric transport conditions.

Specifically, spatial influence between two nodes i and j is adjusted using wind alignment:

- If station j lies downwind of a pollutant source at station i , the edge weight $A_{ij}(t)$ is increased.
- If station j is upwind, the influence is significantly reduced or dampened.
- Wind direction is encoded as a unit vector and projected onto inter-station displacement vectors to estimate directional alignment.

This results in a meteorology-conditioned diffusion graph, where pollutant propagation is no longer symmetric but directionally biased, better reflecting real atmospheric behavior.

Additional features include:

- Temporal lag features (5, 30, 60 minutes).
- Rolling pollutant statistics using sliding windows.
- Weather-conditioned embeddings (humidity, temperature, wind speed).
- Sensor reliability scores derived from anomaly deviation history.

This design ensures that both temporal dynamics and physically meaningful spatial diffusion patterns are captured in the model input space.

4.3 Phase 3: Spatiotemporal Graph-Based Forecasting Model: Instead of treating each monitoring station independently, the system models the environment as a dynamic graph:

$$G = (V, E, A(t))$$

where:

- V : monitoring stations
- E : spatial connectivity based on distance and environmental influence.
- $A(t)$: time-varying adjacency matrix reflecting meteorological conditions.

The proposed model combines:

- Graph Attention Networks (GAT) to learn adaptive spatial dependencies.
- Temporal Convolutional Networks (TCN) to capture multi-scale temporal patterns.
- Weather-aware embedding layers to incorporate atmospheric conditions.

The model outputs multi-horizon forecasts for 5, 30, and 60-minute pollutant concentrations.

4.4 Phase 4: Fault-Tolerant Sensor Recovery Module: To handle missing or unreliable sensor readings, a two-stage recovery approach is used:

- Stage 1: Spatial interpolation (Kriging-based estimation) for short-term sensor gaps using neighboring station correlations.
- Stage 2: Graph-based reconstruction model for longer-term failures using learned spatial-temporal dependencies.

This ensures continuity of predictions even under partial network failure conditions.

4.5 Phase 5: Deployment and Proactive Response System: The deployment phase includes a real-time decision support system designed to simulate proactive industrial responses such as ventilation activation or emission reduction strategies.

This module is implemented as a hybrid rule-based + predictive control system, allowing comparison between reactive and anticipatory intervention strategies.

1. Rule-Based Triggering (Baseline Control System): A deterministic threshold mechanism is defined for immediate safety responses:
 - If $PM_{2.5} > 250 \mu g/m^3$, activate emergency ventilation.
 - If $PM_{2.5} > 150 \mu g/m^3$ for more than 10 minutes, trigger warning state.

This baseline reflects standard industrial safety compliance logic used in monitoring systems.

2. Predictive Trend-Based Control (Proposed Intelligent System): To improve early intervention capability, a predictive control layer is introduced based on forecasted pollution trajectories.

Instead of reacting to current values, the system evaluates:

- Predicted concentration at $t + 5, 30, 60$ minutes.
- Rate of increase (first derivative of pollutant concentration over time).
- Persistence probability of exceeding safety thresholds.

A proactive response is triggered when:

- Forecasted $PM_{2.5}(t + 30)$ exceeds safety thresholds, even if current levels are normal.
- A sustained upward trend indicates likely exceedance within the forecast horizon.

3. Combined Decision Logic: The final system operates as a hybrid controller:

- Rule-based logic ensures safety compliance and fail-safe operation.
- Predictive logic enables early intervention before hazardous buildup occurs.

This dual-layer design allows direct comparison between:

- Reactive industrial safety systems.
- Forecast-driven proactive mitigation systems.

and quantifies the benefit of predictive IoT intelligence in real-world deployment scenarios.

5. Evaluation Framework

The system will be evaluated across three dimensions:

- Forecasting performance: MAE, RMSE, and R^2 for pollutant prediction accuracy
- Early warning capability: Time advantage gained in detecting hazardous pollution events before threshold exceedance.
- System performance: End-to-end latency, throughput (messages per second), and stability under high-frequency streaming conditions.

Robustness will be evaluated by introducing artificial sensor failures and measuring degradation in prediction accuracy under increasing missing-data scenarios (5%, 10%, 20%).

6. Baseline Models

To ensure fair benchmarking, the following models will be implemented:

- Historical Average (HA)
- Linear Regression
- Random Forest Regression
- LSTM-based sequential model

These baselines provide a comparison against both traditional statistical and deep learning approaches.

7. Expected Contributions

This project is expected to deliver the following contributions:

- A real-time distributed IoT pipeline for environmental data streaming using Kafka and Flink.
- A spatiotemporal graph-based air quality forecasting model for industrial environments.
- A robust sensor fault-tolerance framework combining statistical and deep learning approaches.
- A prototype proactive industrial response system for early pollution mitigation.
- An open and reproducible experimental framework for real-time air quality analytics.

8. References

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